SMARTX: Porewater Ammonium

May 2026 Samples

2026-09-03

Contents

##Run Information

```r
cat("Run Information:  ") #lets you know what section you're in
```

```
## Run Information:
```

```r
library(here)
```

```
## here() starts at C:/Users/blakleybd/OneDrive - Smithsonian Institution/Documents/GitHub/GCREW
```

```r
#set the run date & user name
  sample_year <- "2026"
  sample_month <- "May"
  user <- "Brenden Blakley"

#identify the files you want to read in
  #read in as a list to accommodate multiple runs in a month
  NH3_files <- c("Raw Data/20260624_SMARTX_Run1_May2026.csv",
                 "Raw Data/20260626_SMARTX_Run2_May2026.csv",
                 "Raw Data/20260807_SMARTX_Run3_May2026.csv",
                 "Raw Data/20260626_2M_SMARTXrr_May2026.csv"
                 )

# Define the file path for QAQC log file - NO Need to change just check year
  file_path <- here("Porewater QAQC Logs", "GCReW_NH4_QAQC_Log.csv")
  file_path_QAQC <- here("Porewater QAQC Logs", "GCReW_SEAL_Check_QAQC_Log.csv")
  final_path <- "Processed Data/GCReW_SMARTX_Porewater_NH4_202605.csv"

#record any notes about the run or anything other info here:
  run_notes <- "SMARTX_441-80 Not Collected; One PE Check Passes One Fails on 20260626, likely air bubbl

#Add SMARTX Prefix if necessary to samples
  Add_Prefix <- "YES"

  #Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

  cat(run_notes)
```

```
## SMARTX_441-80 Not Collected; One PE Check Passes One Fails on 20260626, likely air bubble
```

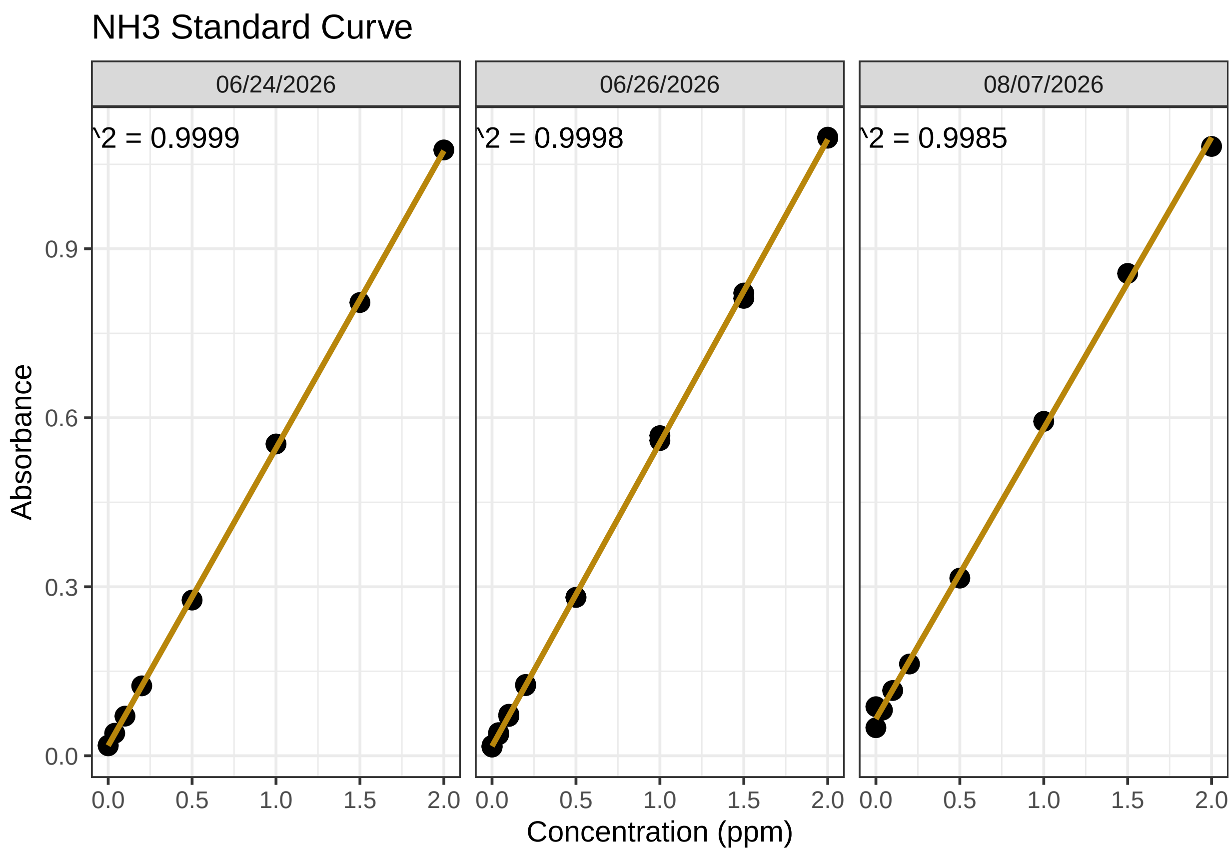
##Setup

##Read in metadata and create similar sample IDs for matching to samples

##Import Data & Clean

0.1 Assessing standard Curves

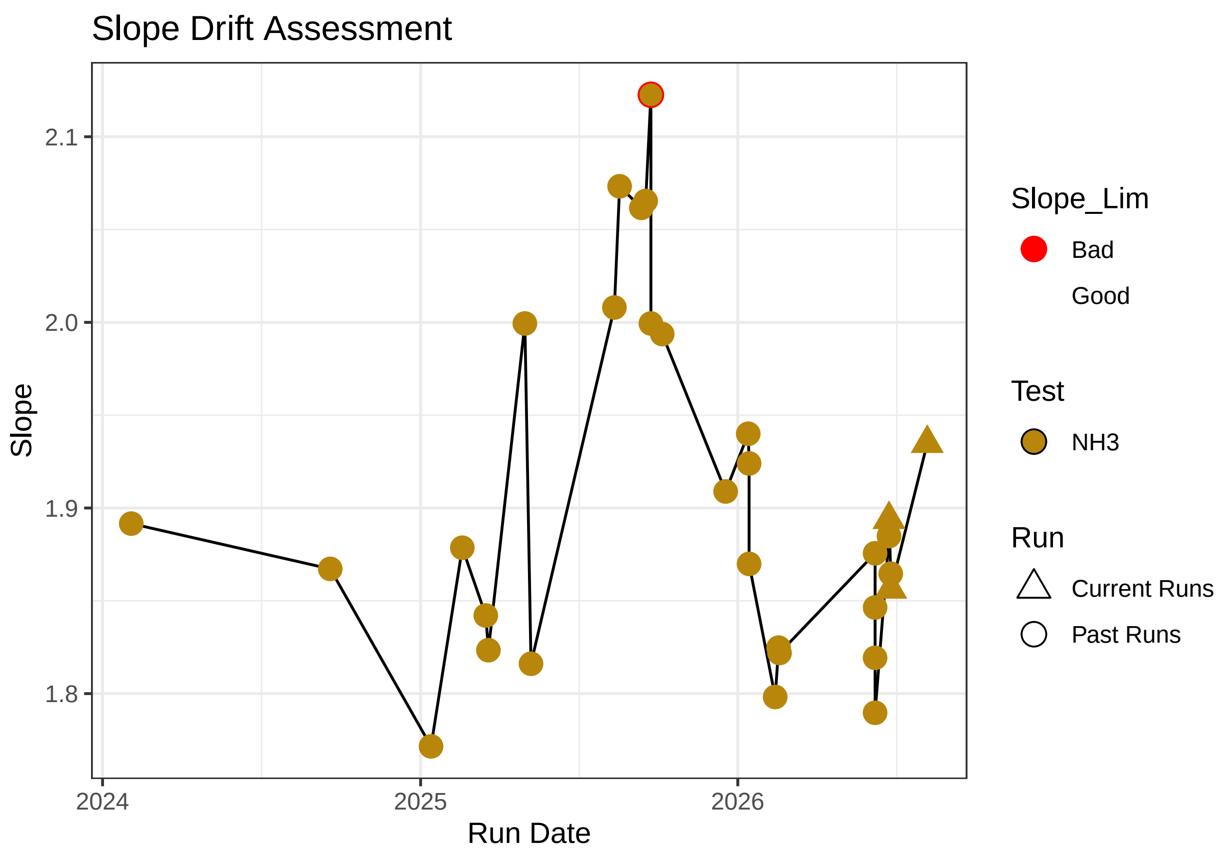
```
## `geom_smooth()` using formula = 'y ~ x'
```



Test	Run_Date	R2_Flag	R2	Slope
Ammonia 2	06/24/2026	R2_Pass	0.9998859	1.893843
Ammonia 2	06/26/2026	R2_Pass	0.9998100	1.856283
Ammonia 2	08/07/2026	R2_Pass	0.9985244	1.934832

0.2 Update Standard Log

```
## `summarise()` has grouped output by 'Test', 'Intercept', 'Slope', 'R_Squared',
## 'User'. You can override using the `.groups` argument.
```


Slope Drift Assessment
Slope
2.1
2.0
1.9
1.8
2024
2025
2026
Run Date
Slope_Lim
Bad
Good
Test
NH3
Run
Current Runs
Past Runs

0.3 Dilution Corrections - ensure the latest dilution is kept

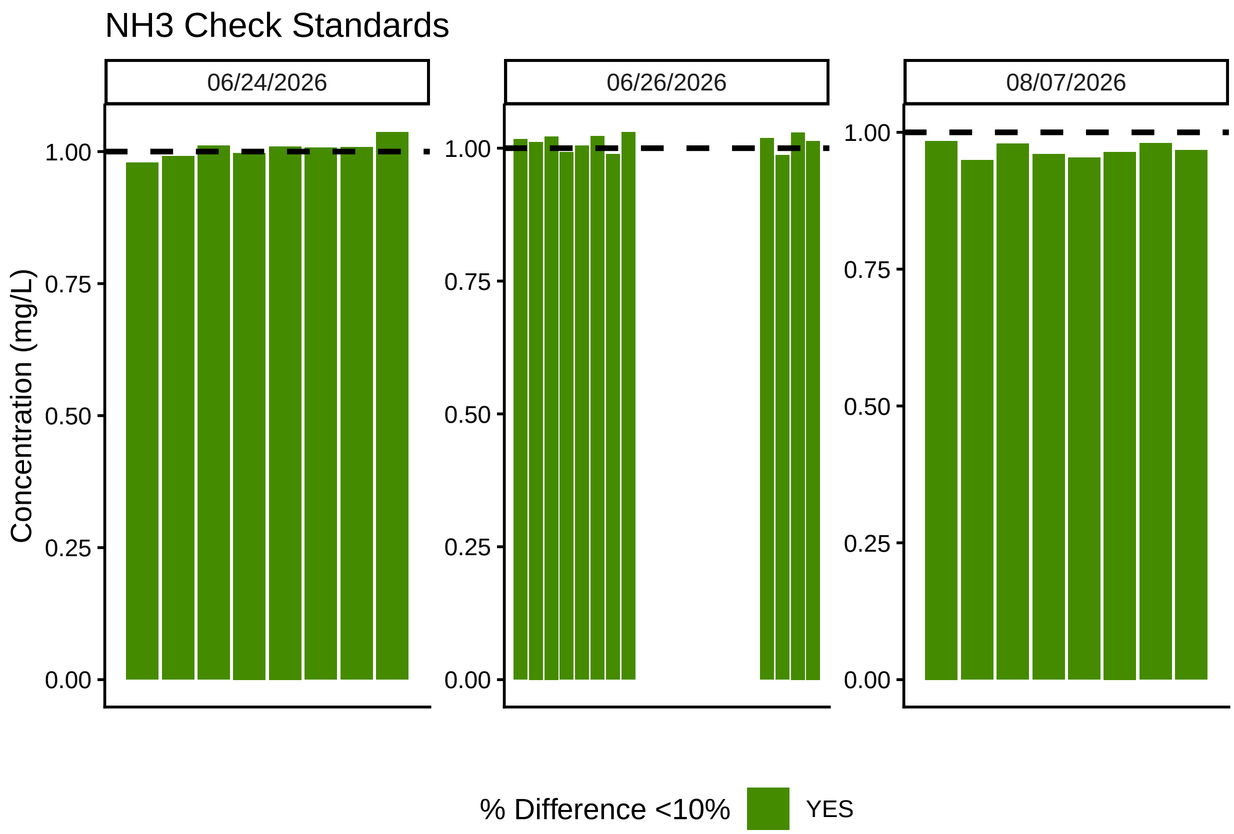
```
## [1] "Reran Samples: c(\"321-20\", \"323-20\", \"331-20\", \"332-20\", \"333-20\", \"633-20\", \"312-

## [1] "Diluted Samples: c(\"321-20\", \"323-20\", \"331-20\", \"332-20\", \"333-20\", \"633-20\", \"31

## [1] "No Naming Issues Detected"
```

0.4 Performance Check

Test	Run_Date	PE_Flag	PE_Conc	PE_Target_Conc
Ammonia 2	06/24/2026	Performance Check Within 25% - PROCEED	1.086850	1.034
Ammonia 2	06/26/2026	Performance Check Within 25% - PROCEED	1.272801	1.034
Ammonia 2	06/26/2026	Performance Check Within 25% - PROCEED	1.272801	1.034
Ammonia 2	08/07/2026	Performance Check Within 25% - PROCEED	1.095402	1.034

0.5 Analyze the Check Standards

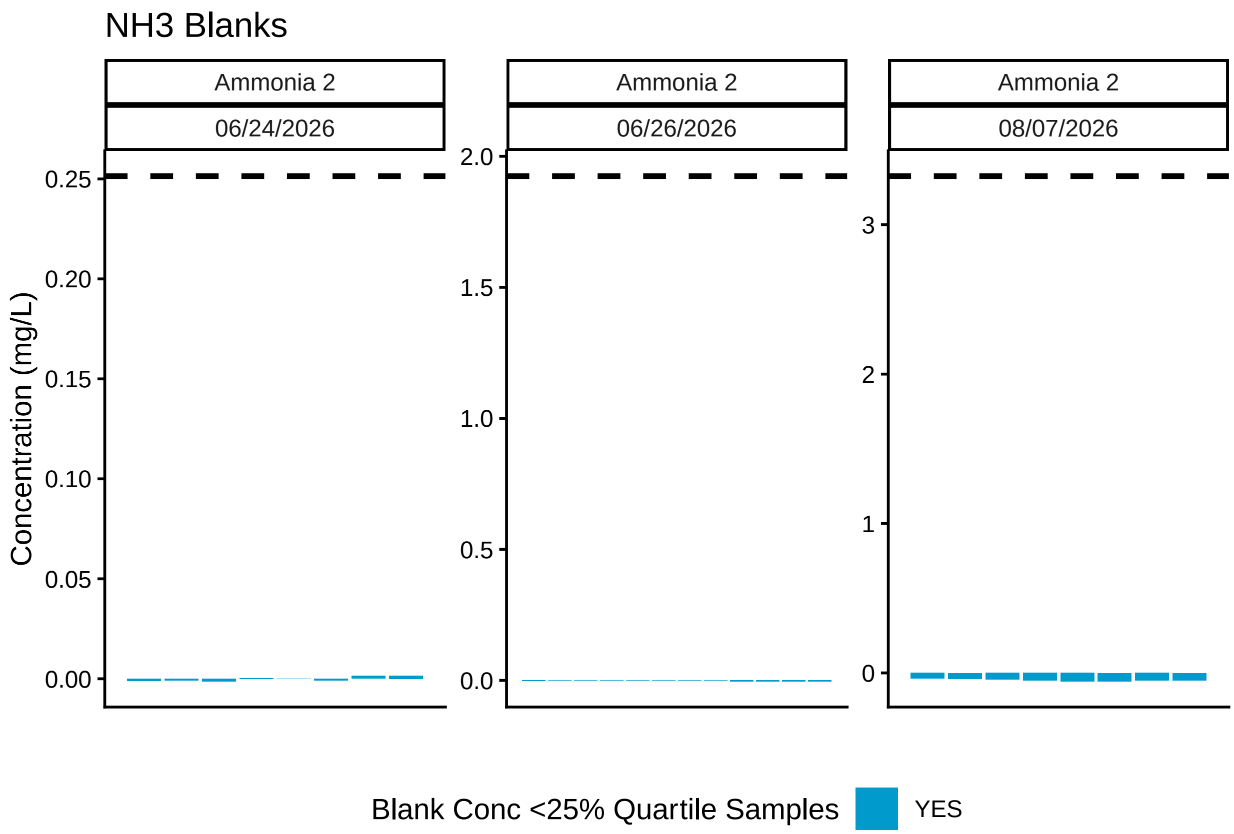
Test	Run_Date	RSV_Flag	RSV	RSV_Cutoff
Ammonia 2	06/24/2026	RSV WITHIN RANGE - PROCEED	0.0242693	0.25
Ammonia 2	06/26/2026	RSV WITHIN RANGE - PROCEED	0.0242693	0.25
Ammonia 2	08/07/2026	RSV WITHIN RANGE - PROCEED	0.0242693	0.25



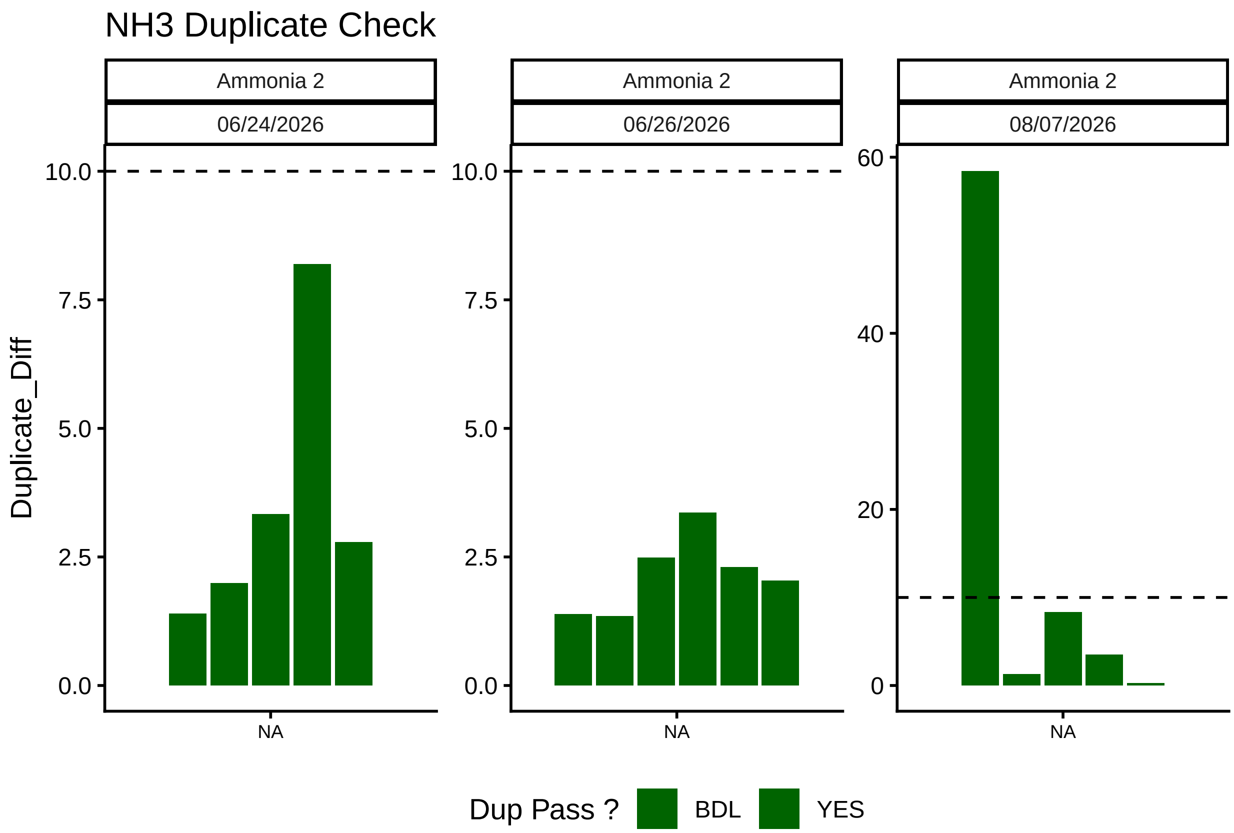
Test	Run_Date	CHK_Flag
Ammonia 2	06/24/2026	>60% of Checks Pass - PROCEED
Ammonia 2	06/26/2026	>60% of Checks Pass - PROCEED
Ammonia 2	08/07/2026	>60% of Checks Pass - PROCEED

0.6 Analyze Blanks

Test	Run_Date	BLK_Pct_Flag	Mean_Blk_Conc	Quantile_25
Ammonia 2	06/24/2026	>60% of Blanks Pass - PROCEED	-0.0001170	0.2514288
Ammonia 2	06/26/2026	>60% of Blanks Pass - PROCEED	-0.0015411	1.9243220
Ammonia 2	08/07/2026	>60% of Blanks Pass - PROCEED	-0.0484810	3.3251115

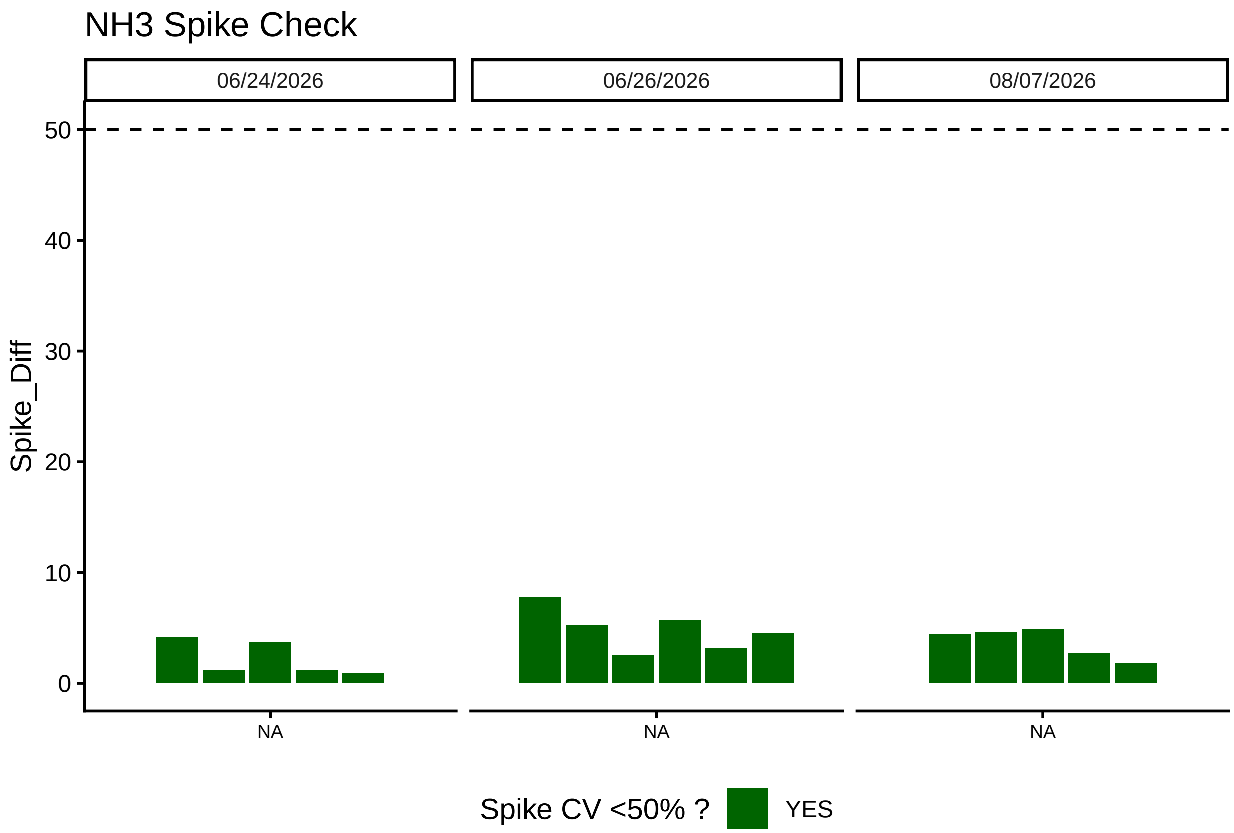


0.7 Analyze Duplicates



Test	Run_Date	Dup_Flags
Ammonia 2	06/24/2026	>60% of Dups Pass - PROCEED
Ammonia 2	06/26/2026	>60% of Dups Pass - PROCEED
Ammonia 2	08/07/2026	>60% of Dups Pass - PROCEED

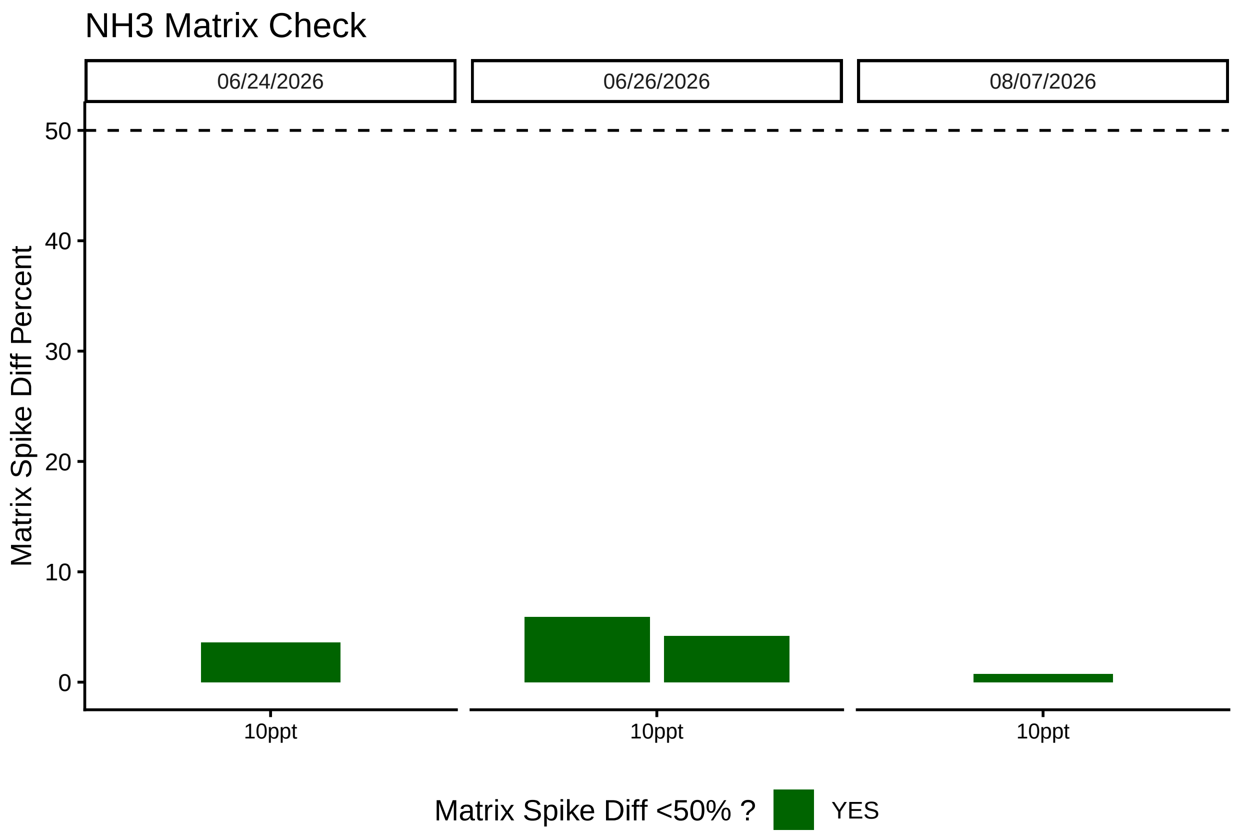
0.8 Spikes



Test	Run_Date	Spike_Flags
Ammonia 2	06/24/2026	>60% of Spikes have a Diff <50% - PROCEED
Ammonia 2	06/26/2026	>60% of Spikes have a Diff <50% - PROCEED
Ammonia 2	08/07/2026	>60% of Spikes have a Diff <50% - PROCEED
NA	NA	>60% of Spikes have a Diff <50% - REASSESS

0.9 Matrix Effects

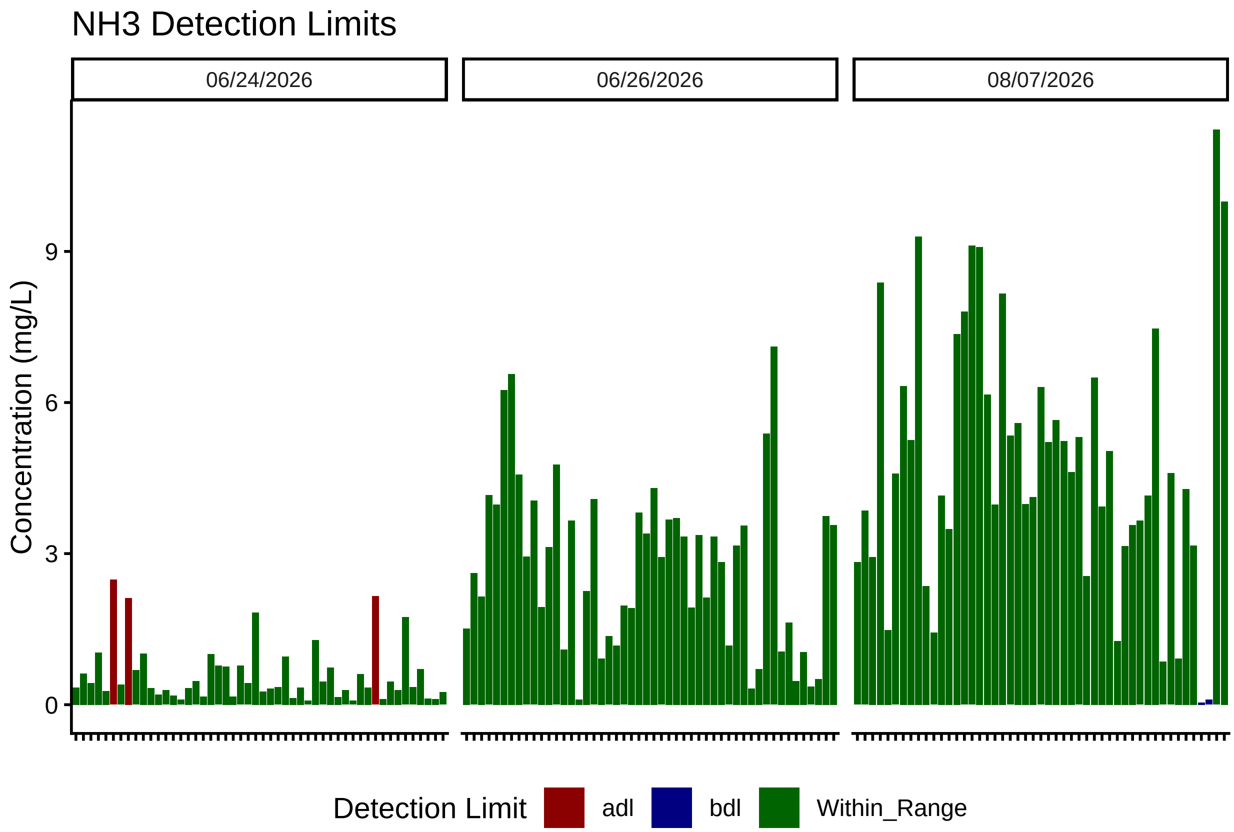
```
## Warning in left_join(Matrix, Spike_ready, by = join_by(Date_Num)): Detected an unexpected many-to-ma
## i Row 7 of ‘x‘ matches multiple rows in ‘y‘.
## i Row 166 of ‘y‘ matches multiple rows in ‘x‘.
## i If a many-to-many relationship is expected, set ‘relationship =
##   "many-to-many"‘ to silence this warning.
```



Test	Run_Date	Matrix_Flags
Ammonia 2	06/24/2026	Matrix Has Diff <50% - PROCEED
Ammonia 2	06/26/2026	Matrix Has Diff <50% - PROCEED
Ammonia 2	08/07/2026	Matrix Has Diff <50% - PROCEED

##Unit Conversion

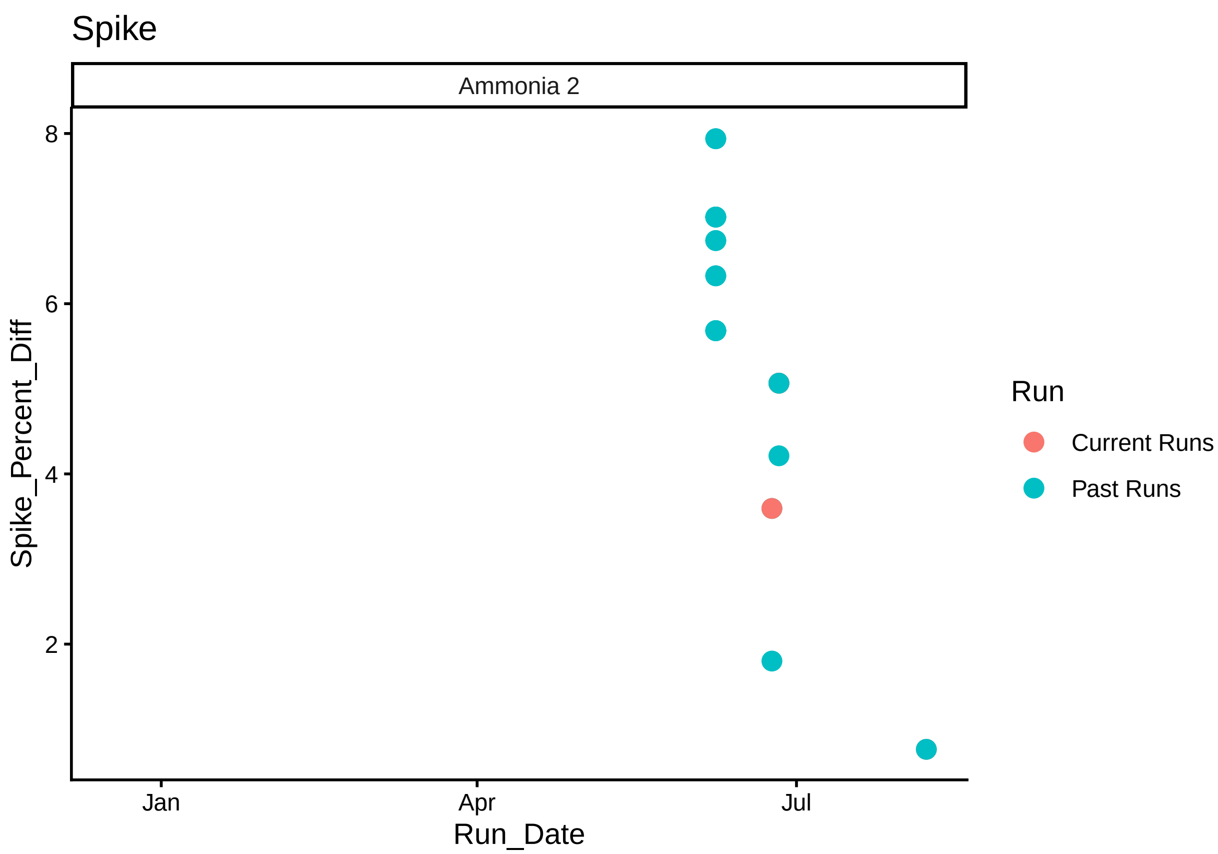
##Sample Flagging - Within range of standard curve

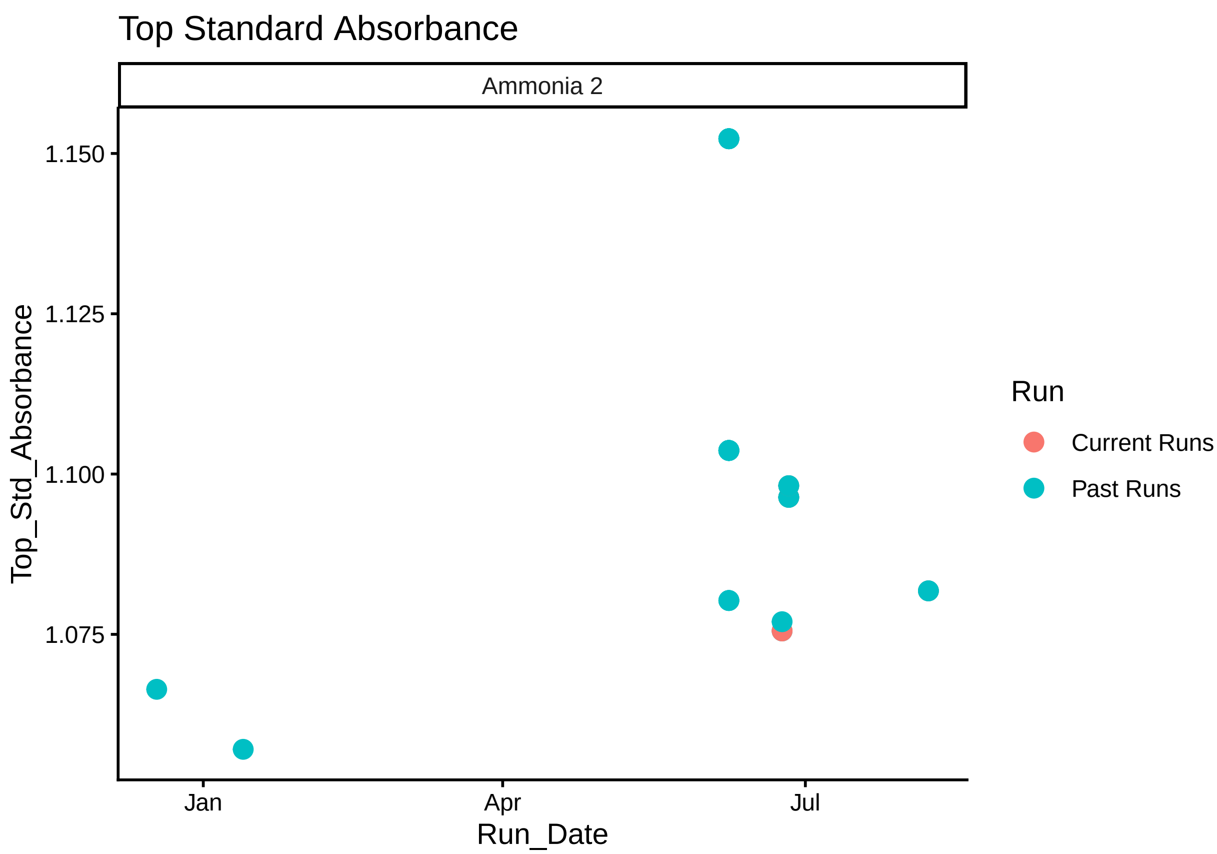


##Add QAQC Flags

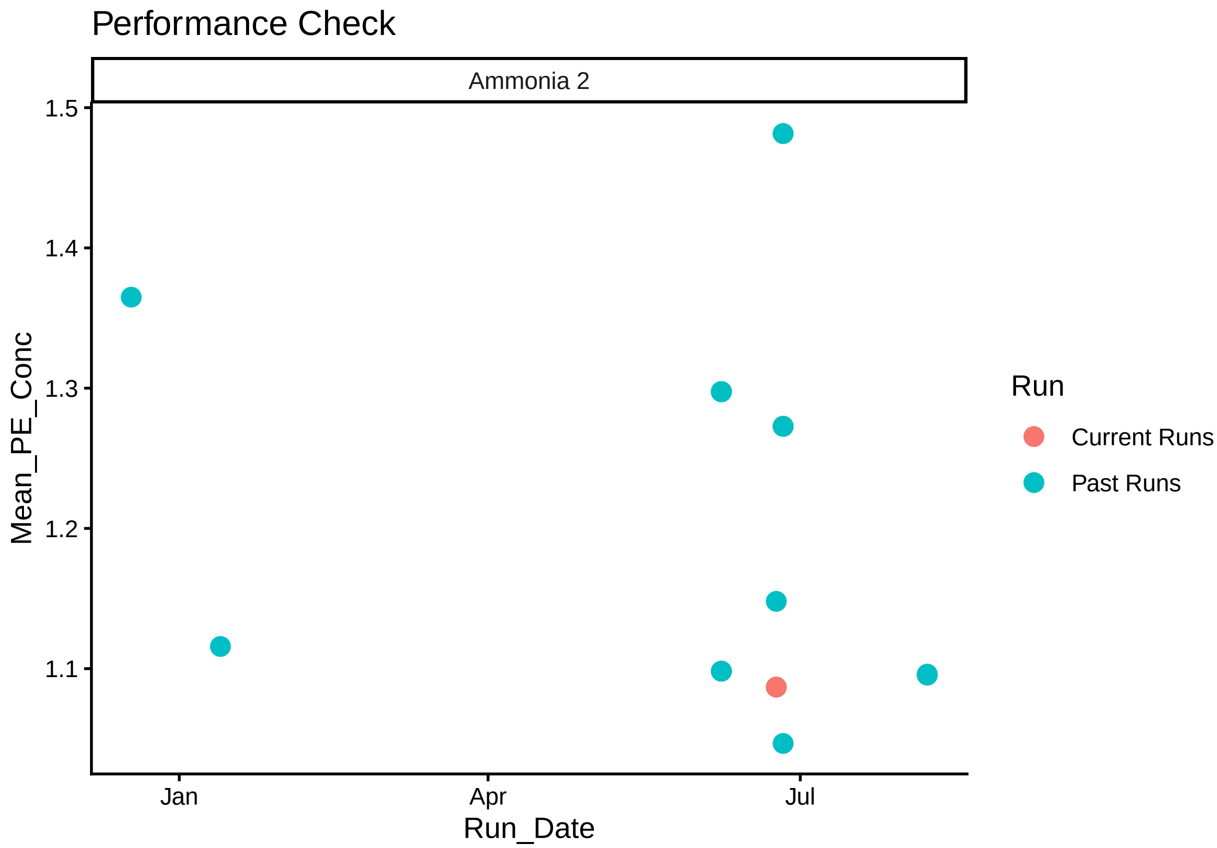
0.10 Create Log for QAQC

```
## Warning in left_join(Top_Standards, PE_Check, by = join_by(Test, Run_Date)): Detected an unexpected m
## i Row 2 of `x` matches multiple rows in `y`.
## i Row 2 of `y` matches multiple rows in `x`.
## i If a many-to-many relationship is expected, set `relationship =
##   "many-to-many"` to silence this warning.

## `summarise()` has grouped output by 'Test', 'Run_Date', 'Top_Std_Absorbance',
## 'Mean_PE_Conc'. You can override using the `.groups` argument.

## Warning: Removed 6 rows containing missing values or values outside the scale range
## (`geom_point()`).
```


Spike
Ammonia 2
Spike_Percent_Diff
8
6
4
2
Jan
Apr
Jul
Run_Date
Run
Current Runs
Past Runs




```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



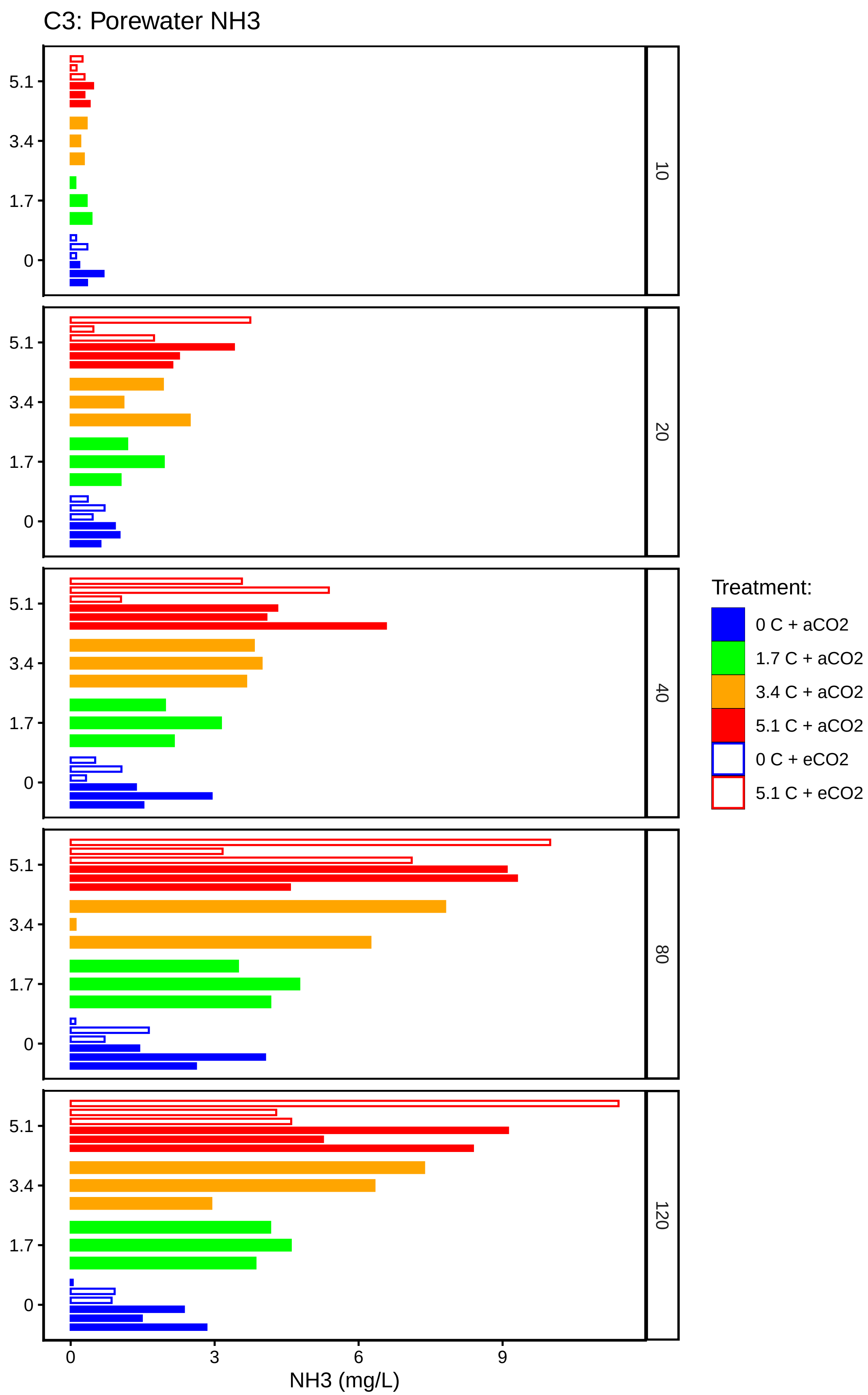
##Write QAQC Log

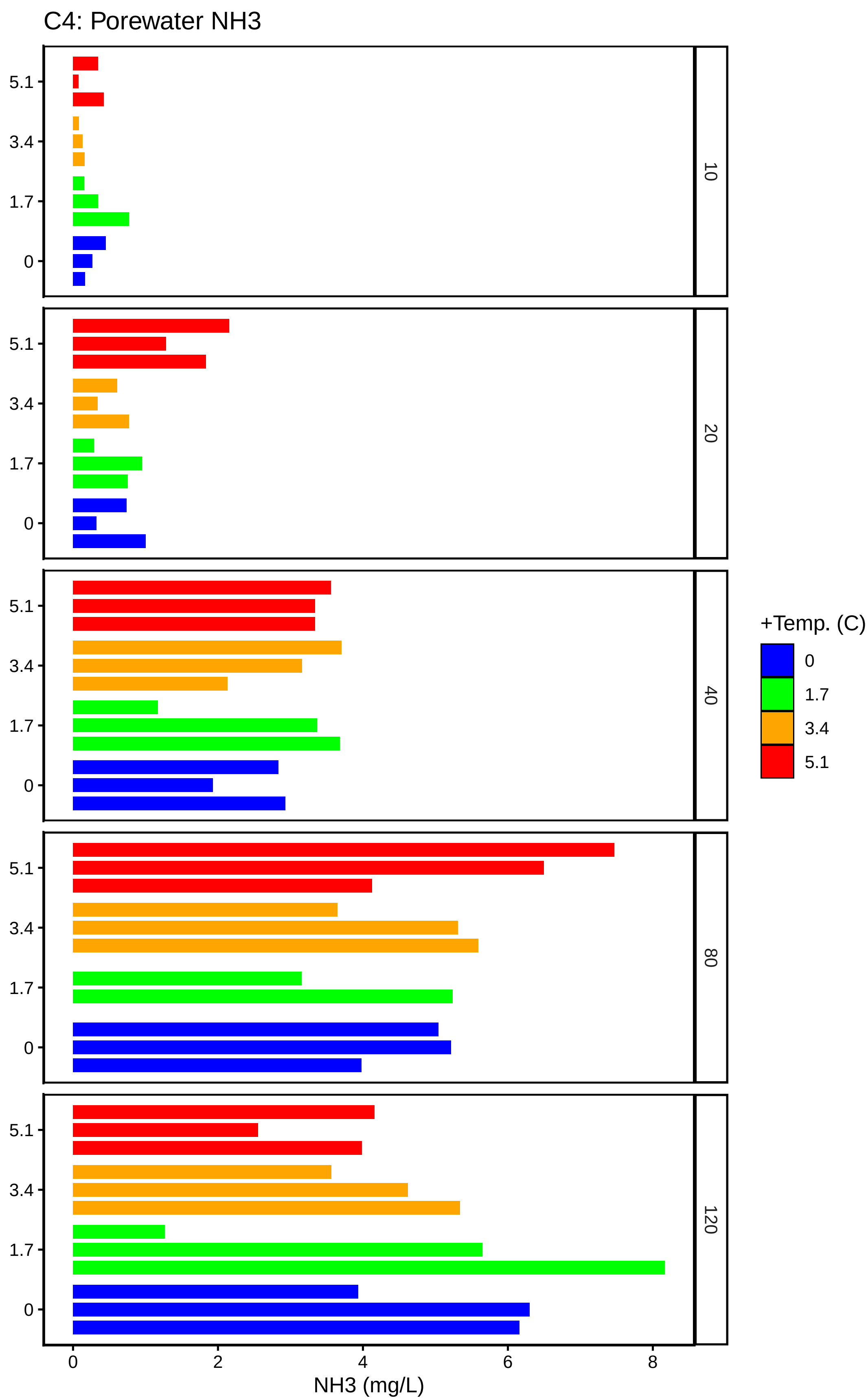
0.11 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata

## Some sample IDs are missing from data.

## [1] "SMARTX_441-80"
```

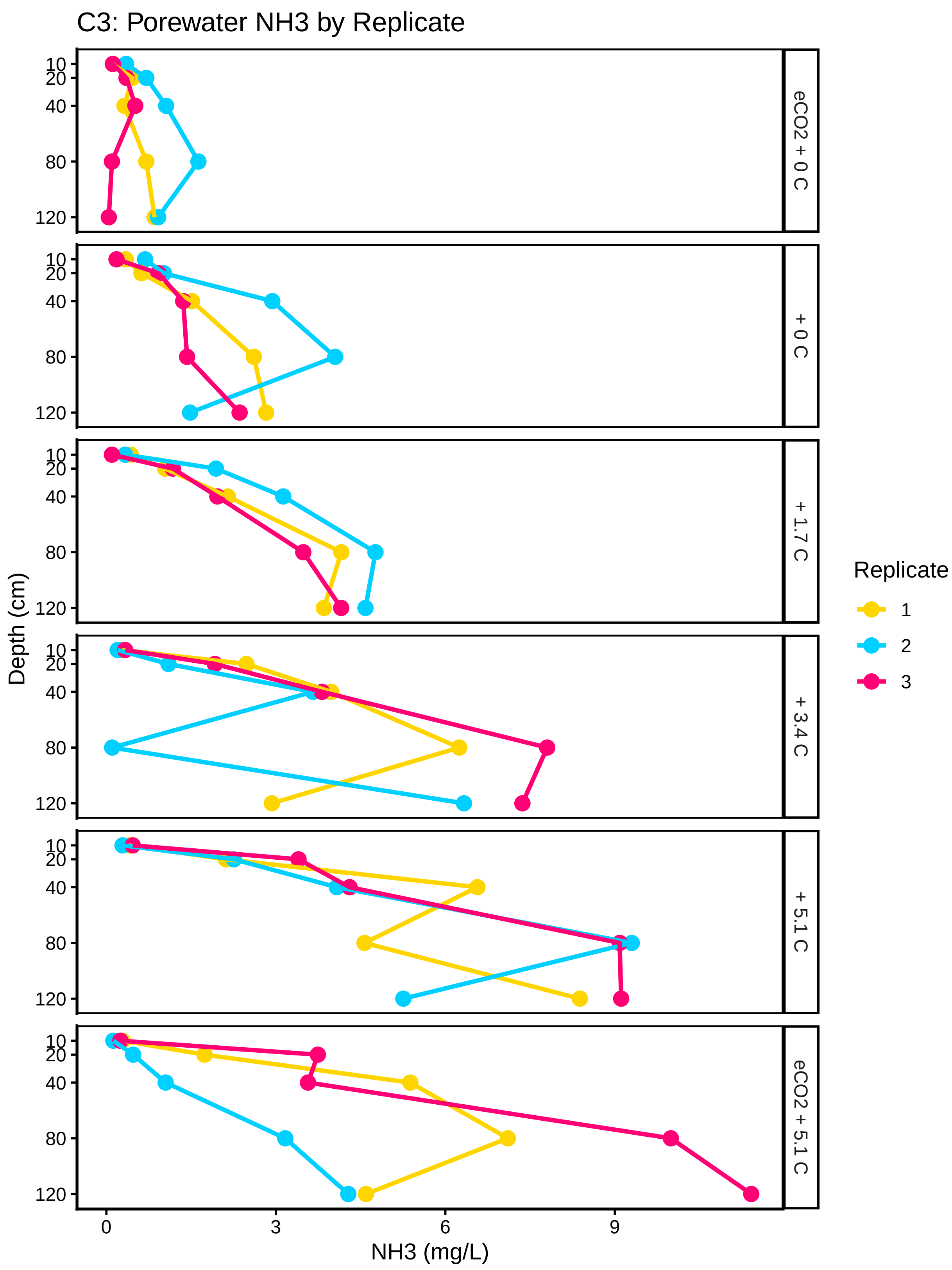
0.12 Visualize Data

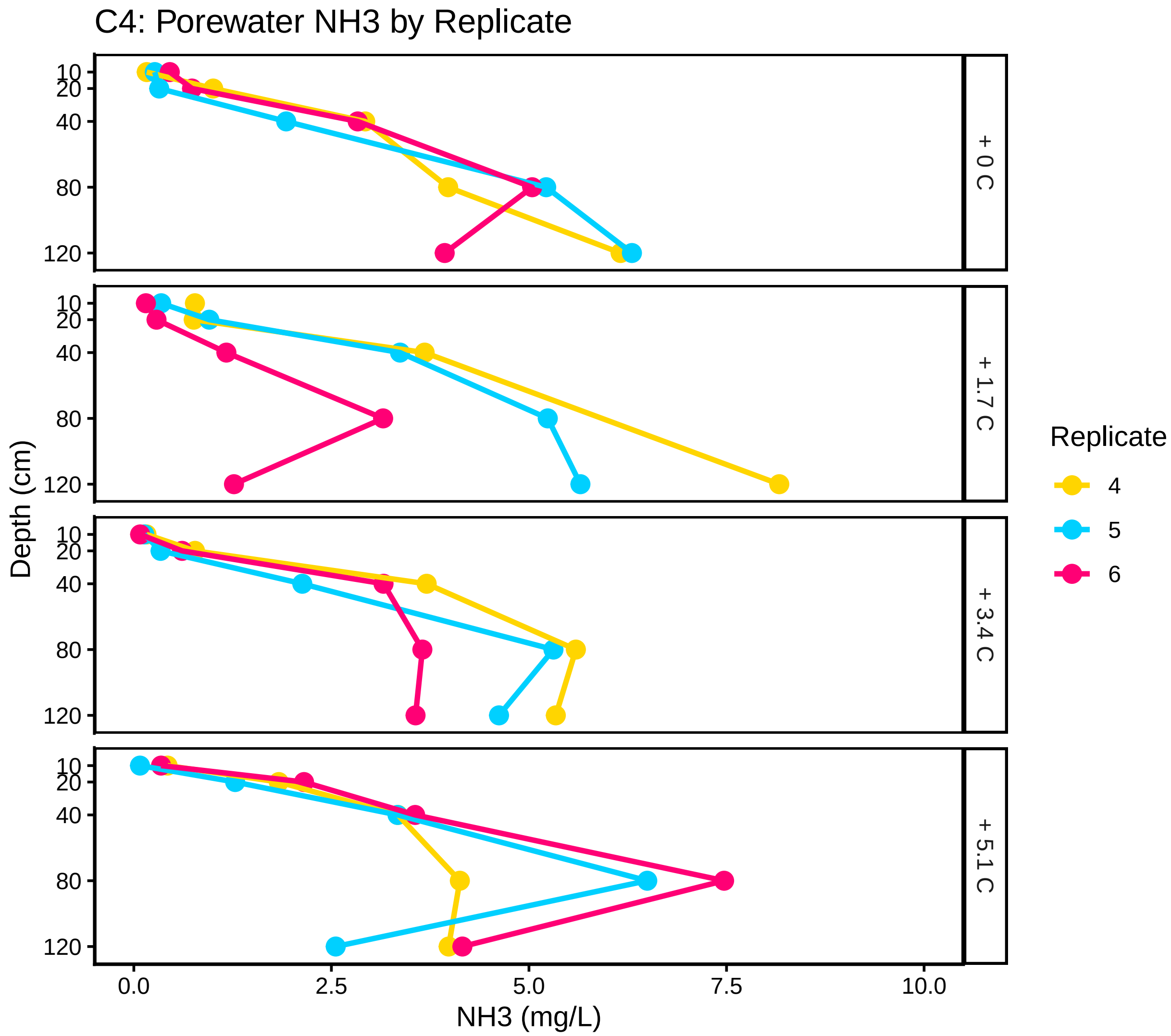
```
## Visualize Data
```


C3: Porewater NH3
5.1
3.4
1.7
0
10
20
40
80
120
0
3
6
9
NH3 (mg/L)
Treatment:
0 C + aCO2
1.7 C + aCO2
3.4 C + aCO2
5.1 C + aCO2
0 C + eCO2
5.1 C + eCO2



C4: Porewater NH3
5.1
3.4
1.7
0
10
20
40
80
120
0
2
4
6
8
NH3 (mg/L)
+Temp. (C)
0
1.7
3.4
5.1


C3: Porewater NH3 by Replicate
Depth (cm)
NH3 (mg/L)
eCO2 + 0 C
+ 0 C
+ 1.7 C
+ 3.4 C
+ 5.1 C
eCO2 + 5.1 C
Replicate
1
2
3
10
20
40
80
120
0
3
6
9



0.13 Organize

#Export Data

#end